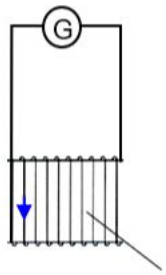
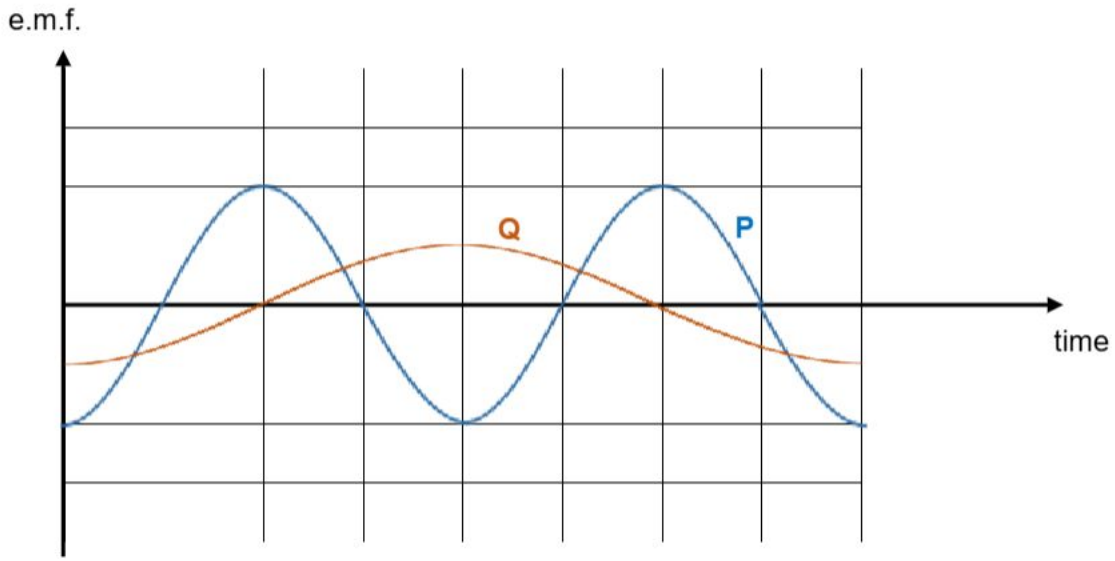
Question 6		
(a)(i)	The e.m.f. induced in a conductor is proportional to the rate of change of magnetic flux linkage. Magnetic flux density generated by solenoid is decreasing (as the current flowing through the solenoid is decreasing). Hence, magnetic flux linkage in small coil is decreasing, so e.m.f. is induced.	**B1** **B1**
(a)(ii)	By Lenz's law, the magnetic field set up in the small coil due to the induced current will be in **the same (rightward) direction as the magnetic field generated by the solenoid** to **oppose the decrease in the flux linkage.** Apply right hand grip rule to determine direction of induced current. Direction of induced current marked correctly on diagram (Induced current should be drawn on a wire and pointing down).  	**B1** **A1**
(b)(i)	 Cosine graph plotted (positive or negative accepted) $I = I_0 \sin\omega t + c$ where $c > 0$ Since $B \propto I$, $\varepsilon = -\frac{d(NBA)}{dt} = -NA\frac{dB}{dt}$ Hence, $\varepsilon \propto -\cos\omega t$	**A1**
(b)(ii)	Peak value halved Period doubled	**A1** **A1**